



**CEP-CCIT**  
**FAKULTAS TEKNIK**

## MINIMARKET POINT OF SALES APPLICATION DATABASE

Group 7

|       |   |                           |              |
|-------|---|---------------------------|--------------|
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Class : 1CC4

CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2024

# **PROJECT ON**

*MINIMARKET POINT OF SALES APPLICATION DATABASE*

## **Developed by**

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## **MINIMARKET POINT OF SALES APPLICATION DATABASE**

Batch Code : 1CC4

Start Date : December 10<sup>th</sup>, 2024

End Date : January 2<sup>th</sup>, 2024

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Date of Submission: January 2<sup>th</sup>, 2024

# CERTIFICATE

This is to certify that this report titled “**Minimarket Point of Sales Application Database**” embodies the original work done by M. Aufar Maulana Prasetia, Briant Abrar Antora, Hilwa Annisa. Project in partial fulfillment of their course requirement at NIIT.

Coordinator:

Muhammad Azka Ramadhan, S.Pd., MBA.

## ACKNOWLEDGEMENT

All praise and gratitude to Allah SWT, the Almighty, for His abundant blessings, guidance, and grace, which have enabled us to successfully complete this Project 4 task, both in the form of this paper and its accompanying presentation, within the given timeframe.

We would like to extend our deepest appreciation to our faculty advisor, Mr. Muhammad Azka Ramadhan, S.Pd., MBA., for his invaluable guidance and continuous support throughout this project. Our gratitude also goes to the entire faculty team, who have been instrumental in enriching our knowledge and providing encouragement during this academic endeavor. We are especially thankful to our classmates for their collaboration, motivation, and friendship, which have been a source of strength in completing this assignment.

This Project 4 paper, titled “Minimarket Point of Sales Application Database”, is submitted as part of the requirements for the Project 4 assignment. Through this paper, we aim to explore and present insights into the development and management of database systems for point-of-sales applications in minimarkets.

We are fully aware that this paper is not without flaws and that there is always room for improvement. Therefore, we sincerely apologize for any mistakes or imperfections that may be found and humbly invite readers to provide constructive criticism and suggestions for betterment.

Lastly, we hope that this paper will serve as a useful resource, not only for academic purposes but also for practical application in the field. We also wish that it inspires further study and development in similar areas. May this report benefit its authors, readers, and the broader academic community.

Depok, December 6<sup>th</sup> 2024

Group 7

# SYSTEM ANALYSIS

## **System Summary:**

The Minimarket Point of Sales (POS) Application Database is a structured system that serves as the backbone for managing the day-to-day operations of a minimarket. It ensures seamless handling of sales transactions, inventory management, user roles, supplier relationships, and reporting functionalities. This database design focuses on efficiency, scalability, and security to meet the needs of both small and growing minimarkets.

# DATABASE DESIGN

Database Name : db\_minimarket

Number of Tables : 6

- tb\_category
- tb\_employeer
- tb\_product
- tb\_supplier
- tb\_transaction
- tb\_trans\_detail

Number of View : 5

- view\_sales\_report
- view\_period\_sales
- view\_low\_stock\_products
- view\_detail\_transaksi
- view\_employee\_performance

Number of Procedure : 5

- proc\_add\_employee
- proc\_add\_new\_product
- proc\_add\_to\_cart
- proc\_new\_transactions
- proc\_update\_stock

# DATABASE DESIGN

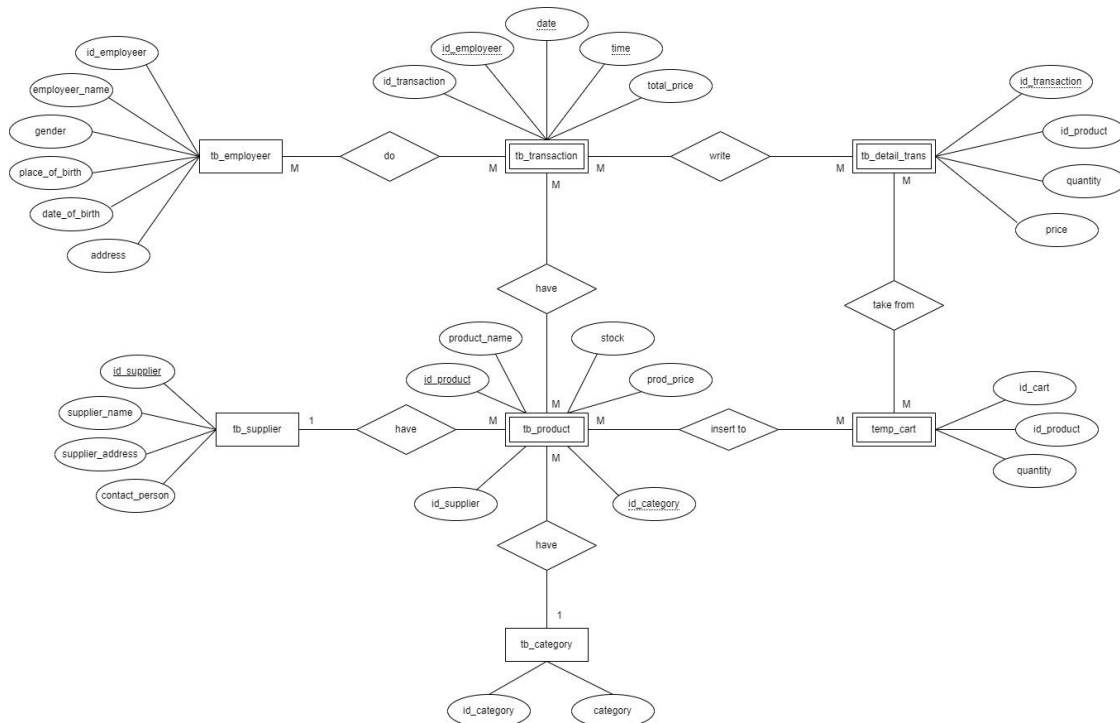
Number of Triggers 5

- trg\_check\_stock
- trg\_before\_insert\_tb\_employe
- trg\_before\_update\_transactio
- trg\_cant\_delete\_trans\_history
- trg\_cant\_same\_name\_product



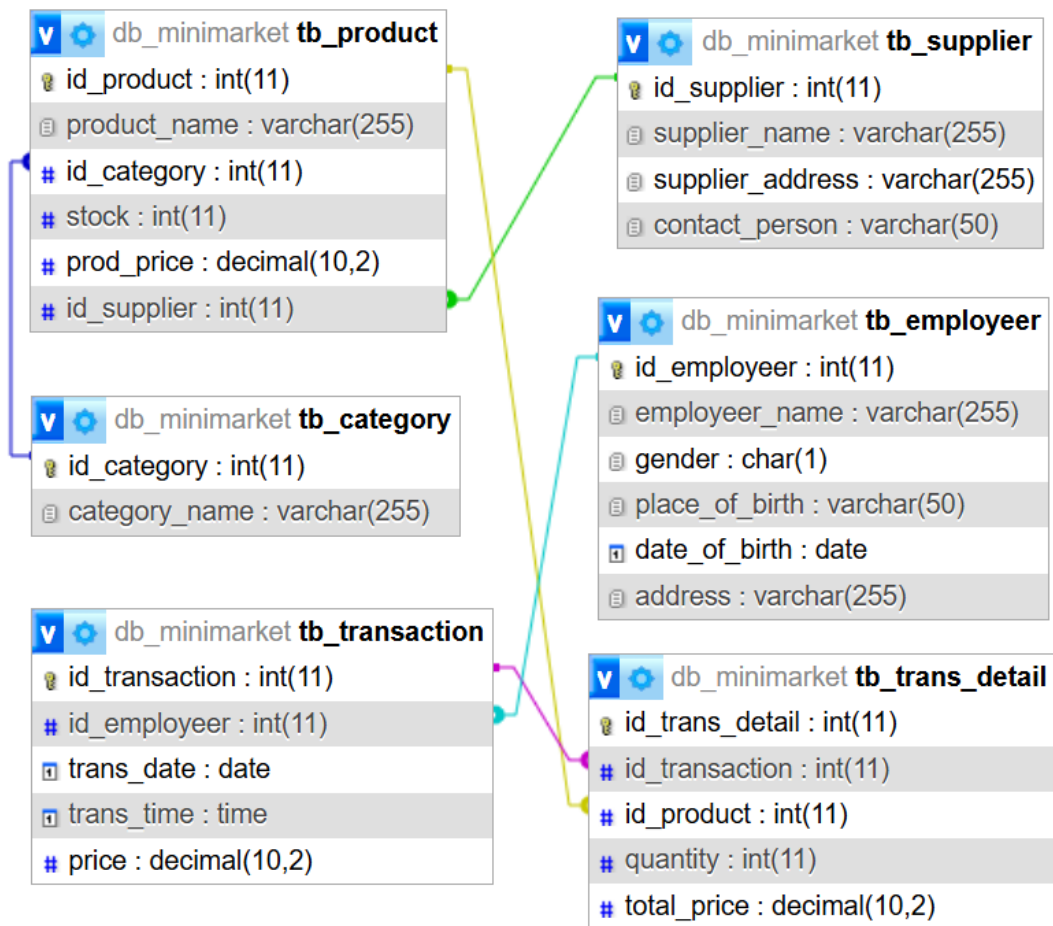
## ERD (Entity Relationship Diagram)

Entity Relationship Diagram, also known as ERD, ER Diagram or ER model, is a type of structural diagram for use in database design. And here is the ERD of the minimarket database.



# SCHEMATIC DIAGRAM

A class diagram is an illustration of the relationships and source code dependencies among classes in the Unified Modeling Language (UML). And here is the Class Diagram of the Minimarket Point of Sales Application Database.



# TABLE DESIGN

The table design of the minimarket database is as follows:

Table category

| #                          | Nama                 | Jenis        | Penyortiran        | Atribut | Tak Ternilai | Bawaan    | Komentar | Ekstra         | Tindakan            |
|----------------------------|----------------------|--------------|--------------------|---------|--------------|-----------|----------|----------------|---------------------|
| <input type="checkbox"/> 1 | <b>id_category</b>   | int(11)      |                    |         | Tidak        | Tidak ada |          | AUTO_INCREMENT | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 2 | <b>category_name</b> | varchar(255) | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |

Table employeeer

| #                          | Nama                   | Jenis        | Penyortiran        | Atribut | Tak Ternilai | Bawaan    | Komentar | Ekstra         | Tindakan            |
|----------------------------|------------------------|--------------|--------------------|---------|--------------|-----------|----------|----------------|---------------------|
| <input type="checkbox"/> 1 | <b>id_employeeer</b>   | int(11)      |                    |         | Tidak        | Tidak ada |          | AUTO_INCREMENT | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 2 | <b>employeeer_name</b> | varchar(255) | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 3 | <b>gender</b>          | char(1)      | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 4 | <b>place_of_birth</b>  | varchar(50)  | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 5 | <b>date_of_birth</b>   | date         |                    |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 6 | <b>address</b>         | varchar(255) | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |

Table products

| #                          | Nama                | Jenis         | Penyortiran        | Atribut | Tak Ternilai | Bawaan    | Komentar | Ekstra         | Tindakan            |
|----------------------------|---------------------|---------------|--------------------|---------|--------------|-----------|----------|----------------|---------------------|
| <input type="checkbox"/> 1 | <b>id_product</b>   | int(11)       |                    |         | Tidak        | Tidak ada |          | AUTO_INCREMENT | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 2 | <b>product_name</b> | varchar(255)  | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 3 | <b>id_category</b>  | int(11)       |                    |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 4 | <b>stock</b>        | int(11)       |                    |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 5 | <b>prod_price</b>   | decimal(10,2) |                    |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 6 | <b>id_supplier</b>  | int(11)       |                    |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |

Table supplier

| #                          | Nama                    | Jenis        | Penyortiran        | Atribut | Tak Ternilai | Bawaan    | Komentar | Ekstra         | Tindakan            |
|----------------------------|-------------------------|--------------|--------------------|---------|--------------|-----------|----------|----------------|---------------------|
| <input type="checkbox"/> 1 | <b>id_supplier</b>      | int(11)      |                    |         | Tidak        | Tidak ada |          | AUTO_INCREMENT | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 2 | <b>supplier_name</b>    | varchar(255) | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 3 | <b>supplier_address</b> | varchar(255) | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 4 | <b>contact_person</b>   | varchar(50)  | utf8mb4_general_ci |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |

# TABLE DESIGN

Table transaction

| #                          | Nama                  | Jenis         | Penyortiran | Atribut | Tak Ternilai | Bawaan    | Komentar | Ekstra         | Tindakan            |
|----------------------------|-----------------------|---------------|-------------|---------|--------------|-----------|----------|----------------|---------------------|
| <input type="checkbox"/> 1 | <b>id_transaction</b> | int(11)       |             |         | Tidak        | Tidak ada |          | AUTO_INCREMENT | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 2 | <b>id_employeer</b>   | int(11)       |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 3 | <b>trans_date</b>     | date          |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 4 | <b>trans_time</b>     | time          |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 5 | <b>price</b>          | decimal(10,2) |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |

Table transaction detail

| #                          | Nama                   | Jenis         | Penyortiran | Atribut | Tak Ternilai | Bawaan    | Komentar | Ekstra         | Tindakan            |
|----------------------------|------------------------|---------------|-------------|---------|--------------|-----------|----------|----------------|---------------------|
| <input type="checkbox"/> 1 | <b>id_trans_detail</b> | int(11)       |             |         | Tidak        | Tidak ada |          | AUTO_INCREMENT | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 2 | <b>id_transaction</b>  | int(11)       |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 3 | <b>id_product</b>      | int(11)       |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 4 | <b>quantity</b>        | int(11)       |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |
| <input type="checkbox"/> 5 | <b>total_price</b>     | decimal(10,2) |             |         | Tidak        | Tidak ada |          |                | Ubah  Hapus Lainnya |

# VALIDATION PERFORM

Category Table

| Column                    | Method(s) user for Validation | Description                                             |
|---------------------------|-------------------------------|---------------------------------------------------------|
| id_category ( <b>PK</b> ) | Primary key, <b>NOT NULL</b>  | <b>This column contain Primary Key, cannot be empty</b> |
| category_name             | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                      |

Employee Table

| Column                    | Method(s) user for Validation | Description                                                         |
|---------------------------|-------------------------------|---------------------------------------------------------------------|
| id_employee ( <b>PK</b> ) | Primary Key, <b>NOT NULL</b>  | <b>This column contain Primary Key, this column cannot be empty</b> |
| employee_name             | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                  |
| gender                    | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                  |
| place_of_birth            | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                  |
| date_of_birth             | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                  |
| address                   | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                  |

Products Table

| Column                    | Method(s) user for Validation | Description                                                                            |
|---------------------------|-------------------------------|----------------------------------------------------------------------------------------|
| Id_product ( <b>PK</b> )  | Primary Key, <b>NOT NULL</b>  | <b>This column contain Primary Key, This column cannot be empty</b>                    |
| product_name              | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                                     |
| Id_category ( <b>FK</b> ) | Foreign Key, <b>NOT NULL</b>  | <b>This column contain Foreign Key, references from category table cannot be empty</b> |
| stock                     | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                                     |
| prod_price                | <b>NOT NULL</b>               | <b>This column cannot be empty</b>                                                     |
| id_supplier ( <b>FK</b> ) | Foreign Key, <b>NOT NULL</b>  | <b>This column contain Foreign Key, references from supplier table cannot be empty</b> |

## VALIDATION PERFORM

Supplier Table

| Column                  | Method(s) user for Validation | Description                                             |
|-------------------------|-------------------------------|---------------------------------------------------------|
| id_supplier <b>(PK)</b> | Primary Key, NOT NULL         | <b>This column contain Primary Key, cannot be empty</b> |
| supplier_name           | NOT NULL                      | <b>This column cannot be empty</b>                      |
| supplier_address        | NOT NULL                      | <b>This column cannot be empty</b>                      |
| contact_person          | NOT NULL                      | <b>This column cannot be empty</b>                      |

Transaction Table

| Column                     | Method(s) user for Validation | Description                                             |
|----------------------------|-------------------------------|---------------------------------------------------------|
| id_transaction <b>(PK)</b> | Primary Key, NOT NULL         | <b>This column contain Primary Key, cannot be empty</b> |
| id_employeeer              | NOT NULL                      | <b>This column cannot be empty</b>                      |
| trans_date                 | NOT NULL                      | <b>This column cannot be empty</b>                      |
| trans_time                 | NOT NULL                      | <b>This column cannot be empty</b>                      |
| price                      | NOT NULL                      | <b>This column cannot be empty</b>                      |

Detail Transaction Table

| Column                      | Method(s) user for Validation | Description                                                                               |
|-----------------------------|-------------------------------|-------------------------------------------------------------------------------------------|
| id_trans_detail <b>(PK)</b> | Primary Key, NOT NULL         | <b>This column contain Primary Key, cannot be empty</b>                                   |
| id_transaction <b>(FK)</b>  | Foreign Key, NOT NULL         | <b>This column contain Foreign Key, references from transaction table cannot be empty</b> |
| Id_product <b>(FK)</b>      | Foreign Key, NOT NULL         | <b>This column contain Foreign Key, references from product table cannot be empty</b>     |
| quantity                    | NOT NULL                      | <b>This column cannot be empty</b>                                                        |
| total_price                 | NOT NULL                      | <b>This column cannot be empty</b>                                                        |

# STORED PROCEDURE

## Procedure add\_employee

```
1 BEGIN
2     -- Menambahkan data karyawan baru, abaikan jika ada duplikat
3     INSERT IGNORE INTO tb_employeeer (employeeer_name, gender, place_of_birth, date_of_birth,
4     VALUES (p_employeeer_name, p_gender, p_place_of_birth, p_date_of_birth, p_address);
5
6     -- Menampilkan pesan sukses
7     -- Perhatikan bahwa pesan ini akan selalu muncul,
8     -- bahkan jika data gagal ditambahkan karena duplikat
9     SELECT 'Data karyawan berhasil ditambahkan.' AS message;
10 END
```

✓ Menampilkan baris 0 - 0 (total 1, Pencarian dilakukan dalam 0,0178 detik.)

`call add_employee ('zara', 'F', 'Kalimantan', '1999-01-01', 'Jl.Kober No. 1');`

[ Edit dikotak ] [ Ubah ] [ Buat kode PHP ]

☐ Tampilkan semua | Jumlah baris: 25 ▾ Saring baris: Cari di tabel ini

Extra options

**message**

Data karyawan berhasil ditambahkan.

# STORED PROCEDURE

## Procedure add\_new\_product

```
1 BEGIN
2     -- Insert data produk baru (tanpa menentukan id_product)
3     INSERT INTO tb_product (product_name, id_category, stock, prod_price, id_supplier)
4     VALUES (p_product_name, p_id_category, p_stock, p_prod_price, p_id_supplier);
5
6     -- Menampilkan pesan sukses dengan ID produk (opsional)
7     -- SELECT LAST_INSERT_ID();
8 END
```

✓ MySQL memberikan hasil kosong (atau nol baris). (Pencarian dilakukan dalam 0,0086 detik.)

```
call add_new_product ('1056', 'Cimory', '1', '76', '13000', '1');
```

[ Edit dikotak ] [ Ubah ] [ Buat kode PHP ]

☐ Show all | Number of rows: 25 | Filter rows: Search in this table | Sort by key: There isn't any

Extra options

| <div><div><div><div></div><div></div></div><div><div></div><div></div></div></div><div></div></div>                                      |      |                   |   | id_product | product_name | id_category | stock | product_price | id_supplier |
|------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|---|------------|--------------|-------------|-------|---------------|-------------|
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 1    | Aqua              | 1 | 50         | 5000.00      | 1           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 2    | Sosro Bottled Tea | 1 | 47         | 4500.00      | 1           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 3    | Ultra Milk        | 1 | 3          | 6500.00      | 1           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 4    | KokoKrunch        | 2 | 64         | 13000.00     | 1           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 5    | Lays              | 2 | 6          | 12000.00     | 2           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 6    | Wheat Essence     | 2 | 59         | 9900.00      | 2           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 7    | KitKat            | 2 | 1          | 18000.00     | 2           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 8    | Silver Queen      | 2 | 57         | 18500.00     | 2           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 9    | Chitato           | 2 | 69         | 15000.00     | 2           |       |               |             |
| <div><div><div></div><div></div></div><div><div></div><div></div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 1056 | Cimory            | 1 | 76         | 13000.00     | 1           |       |               |             |



# STORED PROCEDURE

Procedure add\_to\_cart

```
1 BEGIN
2     INSERT INTO temp_cart (id_product, quantity)
3     VALUES (p_id_product, p_quantity);
4 END
```

✓ MySQL memberikan hasil kosong (atau nol baris). (Pencarian dilakukan dalam 0,0227 detik.)

`call add_to_cart ('1', '2');`

[\[ Edit dikotak \]](#) [\[ Ubah \]](#) [\[ Buat kode PHP \]](#)

✓ Menampilkan baris 0 - 0 (total 1, Pencarian dilakukan dalam 0,0007 detik.)

`SELECT * FROM `temp_cart``

☐ Profil [\[ Edit dikotak \]](#) [\[ Ubah \]](#) [\[ Jelaskan SQL \]](#) [\[ Buat kode PHP \]](#) [\[ Segarkan \]](#)

☐ Tampilkan semua | Jumlah baris: 25 ▾ Saring baris:

Extra options

| ← T →                    |                                                                                     |      |                                                                                     | id_cart | id_product                                                                          | quantity |
|--------------------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|----------|
| <input type="checkbox"/> |  | Ubah |  | Salin   |  | Hapus    |
|                          |                                                                                     |      |                                                                                     | 12      | 1                                                                                   | 2        |

# STORED PROCEDURE

Procedure new\_transactions

```
1 BEGIN
2     -- Memasukkan data ke tabel tb_transaction dengan total harga dari temp_cart
3     INSERT INTO tb_transaction (id_employeer, trans_date, trans_time, price)
4     SELECT p_id_employeer, p_trans_date, p_trans_time, SUM(tp.quantity * pr.prod_price)
5     FROM temp_cart tp
6     JOIN tb_product pr ON tp.id_product = pr.id_product;
7
8     -- Mendapatkan id_transaction yang baru saja ditambahkan
9     SET @last_id = LAST_INSERT_ID();
10
11    -- Memasukkan data detail transaksi ke tb_trans_detail
12    INSERT INTO tb_trans_detail (id_transaction, id_product, quantity, total_price)
13    SELECT @last_id, tp.id_product, tp.quantity, (tp.quantity * pr.prod_price)
14    FROM temp_cart tp
15    JOIN tb_product pr ON tp.id_product = pr.id_product;
16
17    -- Mengurangi stok produk di tb_product
18    UPDATE tb_product pr
19    JOIN temp_cart tc ON pr.id_product = tc.id_product
20    SET pr.stock = pr.stock - tc.quantity;
21
22    -- Menghapus data dari temp_cart
23    DELETE FROM temp_cart;
24 END
```

✓ Kueri SQL Anda berhasil dieksekusi.

0 baris dipengaruhi oleh pernyataan terakhir dalam prosedur.

```
SET @p0='5'; SET @p1='2024-12-31'; SET @p2='17:57:33'; CALL `new_transactions`(@p0, @p1, @p2);
```

# STORED PROCEDURE

Procedure update\_stock

```
1 BEGIN
2     -- Mengecek apakah jumlah_perubahan positif (menambah) atau negatif (mengurangi)
3     IF jumlah_perubahan > 0 THEN
4         UPDATE tb_product
5         SET stock = stock + jumlah_perubahan
6         WHERE id_product = p_id_product; -- Perbaikan pada klausa WHERE
7     ELSE
8         -- Memastikan stok tidak menjadi negatif
9         UPDATE tb_product
10        SET stock = GREATEST(0, stock + jumlah_perubahan)
11        WHERE id_product = p_id_product; -- Perbaikan pada klausa WHERE
12    END IF;
13 END
```

✓ MySQL memberikan hasil kosong (atau nol baris). (Pencarian dilakukan dalam 0,0355 detik.)

```
call update_stock ('5', '25');
```

Edit dikotak ] [ Ubah ] [ Buat kode PHP ]

**VIEW**

In this database there is also a view named "view\_detail\_transaksi".

✓Displaying rows 0 - 15 (total 16, Search performed in 0.0026 seconds.)

SELECT \* FROM `view\_detail\_transaksi`

☐ Profile [Edit in box](#) [Change](#) [Explain SQL](#) [Create PHP code](#) [Refresh](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

|                          |        |      | id_transaction | trans_date | trans_time | cashier  | product_name | category_name     | quantity | total_price |           |
|--------------------------|--------|------|----------------|------------|------------|----------|--------------|-------------------|----------|-------------|-----------|
| <input type="checkbox"/> | Change | Copy | Wipe           | 1          | 2024-12-31 | 10:21:00 | Andra        | Sosro Bottled Tea | Drink    | 2           | 9000.00   |
| <input type="checkbox"/> | Change | Copy | Wipe           | 5          | 2024-12-31 | 00:00:12 | Andra        | Aqua              | Drink    | 2           | 10000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 6          | 2024-12-31 | 10:28:00 | Bisma        | Aqua              | Drink    | 2           | 10000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 6          | 2024-12-31 | 10:28:00 | Bisma        | Ultra Milk        | Drink    | 10          | 65000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 7          | 2024-12-31 | 00:00:12 | Andra        | Aqua              | Drink    | 10          | 50000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 13         | 2024-12-31 | 00:00:12 | Andra        | Aqua              | Drink    | 10          | 50000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 17         | 2024-12-31 | 00:00:12 | Andra        | Aqua              | Drink    | 10          | 50000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 18         | 0000-00-00 | 10:48:00 | Andra        | Aqua              | Drink    | 20          | 100000.00 |
| <input type="checkbox"/> | Change | Copy | Wipe           | 19         | 2024-12-31 | 00:00:12 | Andra        | Aqua              | Drink    | 10          | 50000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 19         | 2024-12-31 | 00:00:12 | Andra        | Sosro Bottled Tea | Drink    | 20          | 90000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 20         | 2024-12-31 | 10:00:00 | Bisma        | Aqua              | Drink    | 11          | 55000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 21         | 2024-12-31 | 10:00:00 | Bisma        | Aqua              | Drink    | 11          | 55000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 22         | 2024-12-31 | 10:00:00 | Bisma        | Aqua              | Drink    | 11          | 55000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 23         | 2024-12-31 | 10:00:00 | Bisma        | Aqua              | Drink    | 11          | 55000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 4          | 2024-12-31 | 00:00:12 | Paul         | Chitato           | Snack    | 2           | 30000.00  |
| <input type="checkbox"/> | Change | Copy | Wipe           | 5          | 2024-12-31 | 00:00:12 | Andra        | Chitato           | Snack    | 3           | 45000.00  |

Query:

```
CREATE TABLE `view_detail_transaksi` (  
  `id_transaction` int(11)  
, `trans_date` date  
, `trans_time` time  
, `cashier` varchar(255)  
, `product_name` varchar(255)  
, `category_name` varchar(255)  
, `quantity` int(11)  
, `total_price` decimal(10,2)  
);
```

|                                                                                                         |    |            |          | id_transaction | trans_date        | trans_time | cashier | product_name | category_name | quantity | total_price |
|---------------------------------------------------------------------------------------------------------|----|------------|----------|----------------|-------------------|------------|---------|--------------|---------------|----------|-------------|
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 1  | 2024-12-31 | 10:21:00 | Andra          | Sosro Bottled Tea | Drink      | 2       | 9000.00      |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 5  | 2024-12-31 | 00:00:12 | Andra          | Aqua              | Drink      | 2       | 10000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 6  | 2024-12-31 | 10:28:00 | Bisma          | Aqua              | Drink      | 2       | 10000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 6  | 2024-12-31 | 10:28:00 | Bisma          | Ultra Milk        | Drink      | 10      | 65000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 7  | 2024-12-31 | 00:00:12 | Andra          | Aqua              | Drink      | 10      | 50000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 13 | 2024-12-31 | 00:00:12 | Andra          | Aqua              | Drink      | 10      | 50000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 17 | 2024-12-31 | 00:00:12 | Andra          | Aqua              | Drink      | 10      | 50000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 18 | 0000-00-00 | 10:48:00 | Andra          | Aqua              | Drink      | 20      | 100000.00    |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 19 | 2024-12-31 | 00:00:12 | Andra          | Aqua              | Drink      | 10      | 50000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 19 | 2024-12-31 | 00:00:12 | Andra          | Sosro Bottled Tea | Drink      | 20      | 90000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 20 | 2024-12-31 | 10:00:00 | Bisma          | Aqua              | Drink      | 11      | 55000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 21 | 2024-12-31 | 10:00:00 | Bisma          | Aqua              | Drink      | 11      | 55000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 22 | 2024-12-31 | 10:00:00 | Bisma          | Aqua              | Drink      | 11      | 55000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 23 | 2024-12-31 | 10:00:00 | Bisma          | Aqua              | Drink      | 11      | 55000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 4  | 2024-12-31 | 00:00:12 | Paul           | Chitato           | Snack      | 2       | 30000.00     |               |          |             |
| <div><div></div><div></div><div></div></div> <div><div>Change</div><div>Copy</div><div>Wipe</div></div> | 5  | 2024-12-31 | 00:00:12 | Andra          | Chitato           | Snack      | 3       | 45000.00     |               |          |             |

Query:

```
CREATE TABLE `view_detail_transaksi` (  
  `id_transaction` int(11)  
  , `trans_date` date  
  , `trans_time` time  
  , `cashier` varchar(255)  
  , `product_name` varchar(255)  
  , `category_name` varchar(255)  
  , `quantity` int(11)  
  , `total_price` decimal(10,2)  
);
```

# VIEW

In this database there is also a view named "view\_sales\_report".

✓ Displaying rows 0 - 12 (total 13, Search performed in 0.0098 seconds.)

```
SELECT * FROM `view_laporan_penjualan`
```

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☐ Show all | Number of rows:  Filter rows:

Extra options

| id_transaction | trans_date | cashier | products_sold           | total_number_of_items | total price |
|----------------|------------|---------|-------------------------|-----------------------|-------------|
| 1              | 2024-12-31 | Andra   | Sosro Bottled Tea       | 2                     | 9000.00     |
| 4              | 2024-12-31 | Paul    | Chitato                 | 2                     | 30000.00    |
| 5              | 2024-12-31 | Andra   | Aqua, Chitato           | 5                     | 55000.00    |
| 6              | 2024-12-31 | Bisma   | Aqua, Ultra Milk        | 12                    | 75000.00    |
| 7              | 2024-12-31 | Andra   | Aqua                    | 10                    | 50000.00    |
| 13             | 2024-12-31 | Andra   | Aqua                    | 10                    | 50000.00    |
| 17             | 2024-12-31 | Andra   | Aqua                    | 10                    | 50000.00    |
| 18             | 0000-00-00 | Andra   | Aqua                    | 20                    | 100000.00   |
| 19             | 2024-12-31 | Andra   | Aqua, Sosro Bottled Tea | 30                    | 140000.00   |
| 20             | 2024-12-31 | Bisma   | Aqua                    | 11                    | 55000.00    |
| 21             | 2024-12-31 | Bisma   | Aqua                    | 11                    | 55000.00    |
| 22             | 2024-12-31 | Bisma   | Aqua                    | 11                    | 55000.00    |
| 23             | 2024-12-31 | Bisma   | Aqua                    | 11                    | 55000.00    |

Query:

```
CREATE TABLE `view_laporan_penjualan` (  
  `id_transaction` int(11)  
  , `trans_date` date  
  , `cashier` varchar(255)  
  , `produk_terjual` mediumtext  
  , `total_jumlah_barang` decimal(32,0)  
  , `total_harga` decimal(32,2)  
  );
```

# VIEW

In this database there is also a view named "view\_sales\_per\_period".

✓ Displaying rows 0 - 1 (total 2, Search performed in 0.0058 seconds.)

```
SELECT * FROM `view_penjualan_per_periode`
```

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☐ Show all | Number of rows:  Filter rows:

Extra options

| trans_date | total_sales |
|------------|-------------|
| 0000-00-00 | 100000.00   |
| 2024-12-31 | 1069000.00  |

Query:

```
CREATE TABLE `view_penjualan_per_periode` (  
  `trans_date` date  
  , `total_penjualan` decimal(32,2)  
);
```

# VIEW

In this database there is also a view named "view\_performa\_karyawan".

✓ Showing rows 0 - 2 (total 3, Search performed in 0.0073 seconds.)

```
SELECT * FROM `view_performa_karyawan`
```

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☐ Show all | Number of rows: 25  Filter rows:

Extra options

| id_employeer | employee_name | total_sales |
|--------------|---------------|-------------|
| 1            | Andra         | 824000.00   |
| 2            | Bisma         | 315000.00   |
| 4            | Paul          | 30000.00    |

Query:

```
CREATE TABLE `view_performa_karyawan` (  
  `id_employeer` int(11)  
  ,`employeeer_name` varchar(255)  
  ,`total_penjualan` decimal(32,2)  
  
  );
```

# VIEW

In this database there is also a view named "view\_produk\_stok\_rendah".

Displaying rows 0 - 2 (3 total, Search performed in 0.0014 seconds.)

```
SELECT * FROM `view_produk_stok_rendah`
```

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☐ Show all | Number of rows:  Filter rows:

Extra options

|                          |        |      |      | id_product | product_name | category_name | stock |
|--------------------------|--------|------|------|------------|--------------|---------------|-------|
| <input type="checkbox"/> | Change | Copy | Wipe | 3          | Ultra Milk   | Drink         | 3     |
| <input type="checkbox"/> | Change | Copy | Wipe | 5          | Lays         | Snack         | 6     |
| <input type="checkbox"/> | Change | Copy | Wipe | 7          | KitKat       | Snack         | 1     |

Query:

```
CREATE TABLE `view_produk_stok_rendah` (  
  `id_product` int(11)  
  , `product_name` varchar(255)  
  , `category_name` varchar(255)  
  , `stock` int(11)  
);
```



# TRIGGER

## Trigger before insert table employeeer

```
1 BEGIN
2     -- Memeriksa apakah nama karyawan sudah ada
3     DECLARE jumlah_karyawan INT;
4     SELECT COUNT(*) INTO jumlah_karyawan FROM
tb_employeeer WHERE employeeer_name = NEW.employeeer_name;
5
6     IF jumlah_karyawan > 0 THEN
7         -- Menampilkan pesan error jika nama karyawan
sudah ada
8         SIGNAL SQLSTATE '45000'
9         SET MESSAGE_TEXT = 'Nama karyawan sudah ada
dalam database.';
10    END IF;
11 END
```

## Trigger before update transaction

```
1 BEGIN
2     IF EXISTS (SELECT 1 FROM tb_trans_detail WHERE
id_transaction = OLD.id_transaction) THEN
3         SIGNAL SQLSTATE '45000'
4         SET MESSAGE_TEXT = 'Tidak bisa mengubah transaksi
yang sudah memiliki detail transaksi.';
5     END IF;
6 END
```

# TRIGGER

## Trigger can't delete transaction history

```
1 BEGIN
2     IF EXISTS (SELECT 1 FROM tb_trans_detail WHERE
3 id_transaction = OLD.id_transaction) THEN
4         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Tidak
5 bisa menghapus transaksi yang sudah memiliki detail
transaksi.';
6     END IF;
7 END
```

## Trigger can't same name product

```
1 BEGIN
2     IF EXISTS (SELECT 1 FROM tb_product WHERE product_name
3 = NEW.product_name) THEN
4         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Produk
5 dengan nama tersebut sudah ada.';
6     END IF;
7 END
```

## Trigger check stock

```
1 BEGIN
2     IF NEW.stock < 0 THEN
3         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Stok
4 produk tidak mencukupi';
5     END IF;
6 END
```

# CONFIGURATION

**Hardware** : Laptop

**Operating System** : Windows 11

**Software** : Microsoft Word 2019  
Microsoft Power Point 2019  
XAMPP  
Phpmyadmin on Chrome Browser

## PROJECT FILE DETAILS

| No | File Name                                                    | Remarks                                                |
|----|--------------------------------------------------------------|--------------------------------------------------------|
| 1. | Group 7_ Minimarket Point of Sales Application Database.docx | Microsoft Word that contains report                    |
| 2. | Group 7_ Minimarket Point of Sales Application Database.pdf  | PDF that contains report                               |
| 3. | Group 7_ Minimarket Database.pptx                            | Microsoft Power Point that contains presentation slide |
| 4. | Minimarket_db.sql                                            | Database File                                          |